

A Study of Various Open Source Blockchain Platforms

Blockchain platforms are important because they can be used in e-governance, eliminate middle men, and secure the confidence of businesses. Get acquainted with the various blockchain platforms in this interesting article.



The term 'blockchain technology' typically refers to the transparent, trustless, publicly accessible ledger that securely transfers the ownership of units of value using public key encryption and proof of work methods. The technology uses decentralised consensus to maintain the network, which means it is not centrally controlled by a bank, corporation or government. In fact, the larger the network grows and becomes increasingly decentralised, the more secure it becomes. Blockchain technology is picking up at a fast pace nowadays. The technology that emerged in 2009 as the underlying platform for Bitcoin exchange has now evolved into a mainstream technology. It finds application in various fields like healthcare and finance. Many companies venturing into this domain are developing blockchain based

applications that help in making business operations more transparent and efficient.

What exactly is blockchain?

At the most basic level, blockchain is just a chain of blocks, but not in the traditional sense of those words. When we use the words 'block' and 'chain' in this context, we are actually talking about digital information (block) stored in a public database (chain).

Blocks on the blockchain are made up of digital pieces of information and they have three main parts.

Part 1: Blocks store information about the transaction like the date, time and amount involved in the transaction of the recent purchase.

Part 2: These blocks store information about the participants in the transactions. For example, a block for purchase of some items from

Amazon would record the name along with other information to *amazon.com*. Instead of using the actual name, the purchase is recorded without any identifying information using a unique 'digital signature' like a sort of user name.

Part 3: Blocks store information that distinguishes them from other blocks, much like names of people, and each block stores a unique code called a hash that allows every block to be different from the other. Hashes are the cryptographic codes created by special algorithms.

Blockchain variants

Public: The ledgers of these blockchains are visible to everyone on the Internet. They allow anyone to verify and add a block of transactions to the blockchain. Public networks have incentives for people to join and are

